TUTORIAL 1

Complex Numbers and Induction

1.1 Course Logistics

NOTE

The current International Engineering syllabus and official announcements determine the actual class arrangements, submission deadlines, office hours, and assessment rules. In particular, WeBWorK is optional practice and contributes neither to the grade nor to a protective component in the international course.

1.1.1 Access to Tutorials

Tutorials may be held over Zoom, and recordings may be available. The way links and recordings are made available may differ between staff members; they will usually be found on Moodle. Technical problems may prevent a tutorial from being made available. The course team will try to avoid such problems, but students remain responsible for catching up on any material missed because of a technical failure. Follow the current course announcements for the arrangements that apply to your group.

1.1.2 Homework and Assessment

- For a homework sheet with two parts, the first contains two questions marked for submission, to be solved independently and submitted for grading. The second contains additional questions for independent practice, which are not submitted. Full solutions to some of these questions are provided so that you can assess your own work. Check the current assignment instructions for the questions you must actually submit.
- The homework is based on material from the lectures and/or tutorials.
- Submitting work gives you professional feedback. The additional questions help you consolidate the material and practise solving relevant problems. You are strongly encouraged to attempt as many questions as possible.
- One possible publication schedule is Tuesday morning, with submissions accepted until Saturday evening of the following week (a submission window labelled as 12 days). The current course announcements determine the deadlines for the international program.
- Where WeBWorK sheets are provided, they may be released approximately every week or at a longer interval and may be open for two weeks. International Engineering students may use available WeBWorK questions as optional practice. The material may be in Hebrew; WeBWorK scores do not contribute to the international course grade or to any protective component.

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- The final assessment rules are stated in the current syllabus. Homework instructions appear in the relevant course section. Direct questions about grades and examinations to the course lecturer.

1.1.3 Office Hours

- Office hours with the teaching assistant are intended for questions arising from the tutorial material and homework.
- The teaching assistant sets the time, location, and format of office hours. Office hours are not an additional tutorial and do not replace independent study.

Procedures may change; updates will be announced when necessary.

1.1.4 Study Recommendations

Mathematics is a language, and learning it requires practice. This course is an integral (and differential) part of the mathematical foundations that support your later academic studies. The following suggestions are intended to help you learn the material and manage the course.
Scope of this advice: The recommendations in this section do not make the author responsible for readers' study decisions, understanding, or application of the material. Each reader remains responsible for their own learning and use of the information.

- Attempt all questions marked for submission and all practice questions; they have been selected carefully.
- If you fall slightly behind, use the recordings and tutorials to catch up, and check that you can solve the homework questions on the same topic.
- If you fall substantially behind, prioritize the theoretical foundations and basic practice questions, and try to rejoin the class's pace as soon as possible.
- Consider making a ``three magic sheets'' summary: one sheet for definitions, one for propositions and theorems, and one for useful techniques.
- A competitive atmosphere does not help the group or the individual. Please help one another. Group learning and support for classmates can strengthen both learning and resilience.

1.1.5 Clarifications, Personal Recommendations, and Expectations

- Please send messages and questions through the website system at chengadi.science.

1. Logistics requests should include your full name, student ID, and course name. Requests lacking these details will not be processed. If you have received no response within three days, I may have missed your request; please send a reminder through the system and include or quote the original request.
2. Questions about the material: in previous years and semesters, I received a very large number of questions (at least 20 requests per day). Questions may be sent through the website system, subject to the following guidelines:
   (a) First make an independent effort to understand the issue, using notes, books, the internet, and tools such as ChatGPT. Check mathematical claims independently.

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(b) Keep your message short and specific. Include the course name and a question that can be answered in one or two lines. Attach an image of the relevant material where possible, rather than referring only to a particular theorem.

(c) Do not send requests such as ``Is my method correct?'' or ``Check this part.'' I will not grade homework solutions through messages. You can discuss an approach during office hours or obtain feedback through the homework grader.

(d) You may ask for clarification of the wording of a homework question.

(e) I will try to answer within a few days. Messages that are too long, or that require more than a few minutes to read and answer, may not receive a response. Thank you for your understanding.

- I may upload videos of myself solving problems on YouTube. These are an additional resource for students, rather than part of my formal responsibilities; please do not send requests about their production or availability.
- During tutorials, I will not summarize all the theorems and conclusions covered in lectures. I recommend preparing your own summary. Tutorials are intended to practise the lecture material and work on somewhat more demanding questions that encourage thought and independent learning.
- I will also make sure to include basic exercises.
- Time will be set aside for questions during class. I will not repeatedly answer questions that have already been addressed.
- **The lectures determine the required material.**
  It is not possible to cover every topic in the tutorials or practise every type of question on each topic.
- I strongly recommend exploring the material through the resources available to you: books, online sources, earlier summaries (for example, materials from the students' association), office hours, and similar resources. **Each student is responsible for their own learning.**
- Tutorial pace and presentation: I will try to work at a reasonable pace, write clearly, and use headings. Please let me know if the presentation needs adjustment.
- WeBWorK is optional and recommended as additional practice with feedback. **For the International Engineering Program, WeBWorK does not count toward the final grade and does not provide protective credit.**
- I will try to provide examples, proofs, and counterexamples for each topic.
- I will conduct an anonymous mid-semester teaching survey, and I welcome feedback.
- I will take questions at the designated times rather than in the middle of solving an exercise.
- I will try to upload typed tutorial sheets before the tutorials, but they may contain errors or imprecise statements.
  The material presented in the lectures, both in writing and orally, remains authoritative.
- Come prepared: review the definitions and material covered in the lectures before each tutorial.

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MOTIVATION AND GUIDING IDEA

Cartesian form is well suited to addition and comparison of components; polar form is well suited to multiplication, powers, and roots. Before calculating, choose a representation that fits the requested operation. For induction, first choose the precise statement and the first value for which it is supposed to hold.

NOTATION AND CONVENTIONS

For $z = x+iy \in \mathbb{C}$, the symbols $\operatorname{Re} z = x$ and $\operatorname{Im} z = y$ denote real numbers; in particular, the imaginary part is y, not iy. Both z^* and $\overline{z}$ denote the conjugate $x - iy$. The modulus $|z|$ is a nonnegative real number. When $z \neq 0$, $\arg z$ refers to an argument modulo integer multiples of 2π, whereas $\operatorname{Arg} z \in [0, 2\pi)$ is the fixed choice in these notes. The notation $a \mid b$ means that $b = am$ for some $m \in \mathbb{Z}$. The expression $P(n) \Rightarrow P(n+1)$ is a logical implication; it does not by itself establish the base case.

DEFINITION 1.1

Existence, uniqueness, and conditions of use. Every $z \in \mathbb{C}$ has a unique representation $z = x + iy$ with $x, y \in \mathbb{R}$. Thus two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal. For $z \neq 0$, its inverse exists and is unique, since $zz^* = |z|^2 > 0$; check this condition before dividing by z. If $r > 0$ and $\theta \in \mathbb{R}$, de Moivre's formula gives

$$(re^{i\theta})^n = r^n e^{in\theta} \qquad (n \in \mathbb{Z}).$$

Negative powers require $r > 0$. For nonnegative integer powers, the formula also holds when $r = 0$, except for 0^0, which is not defined here. When finding nth roots of a nonzero number, for $n \geq 1$, list all n roots rather than selecting a single angle.

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How to use these notes

These notes are intended for students in all engineering disciplines. They combine definitions, guiding ideas, examples, and detailed solutions.

MOTIVATION AND GUIDING IDEA

Objectives. We will convert between Cartesian and polar forms, calculate powers and roots, and interpret operations in the complex plane. We will then formulate an induction proof, stating the domain on which its claim holds.

MOTIVATION AND GUIDING IDEA

A working habit. In every solution, state what is given, what must be established, which definition or result you use, and why every operation is permitted. Correct calculations and clear explanations complement one another.

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1.2 Complex numbers: Cartesian form

DEFINITION 1.2

The imaginary unit i satisfies $i^2 = -1$.
A complex number has the form $z = x + iy$, where $x, y \in \mathbb{R}$.

1. If $\mathrm{Re}(z) = x = 0$, then z is purely imaginary.
2. If $\mathrm{Im}(z) = y = 0$, then z is real.

EXERCISE 1.1

Please solve $z^2 + 2z + 17 = 0$ over $\mathbb{C}$. Please factor it into linear factors over $\mathbb{C}$ and please determine whether such a factorization exists over $\mathbb{R}$.

WORKED SOLUTION

The discriminant is $\Delta = 2^2 - 4 \cdot 17 = -64$. Hence

$$z_{1,2} = \frac{-2 \pm \sqrt{-64}}{2} = -1 \pm 4i, \qquad z^2 + 2z + 17 = (z + 1 - 4i)(z + 1 + 4i).$$

The negative discriminant gives no real root, so there is no factorization into two real linear factors.

DEFINITION 1.3

- **Complex conjugate:** for $z = x + iy$, write $z^* := x - iy$.
- **Modulus:** $|z| = \sqrt{x^2 + y^2} \in \mathbb{R}$.
- **Multiplicative inverse:** for $z \neq 0$, $z^{-1} = \dfrac{z^*}{|z|^2}$.

EXERCISE 1.2

Please prove the following statements. **(a)** $\mathrm{Re}\, z = \dfrac{z + z^*}{2}$ and $\mathrm{Im}\, z = \dfrac{z - z^*}{2i}$. **(b)** $|z| \geq \mathrm{Im}\, z$ and $|z| \geq \mathrm{Re}\, z$ for every $z \in \mathbb{C}$. **(c)** (At home) $(z^*)^* = z$. **(d)** (At home) If z is real, then $z^* = z$.

WORKED SOLUTION

Write $z = x + iy$. Then

$$\frac{z + z^*}{2} = x = \mathrm{Re}\, z, \qquad \frac{z - z^*}{2i} = y = \mathrm{Im}\, z,$$

and

$$|z| = \sqrt{x^2 + y^2} \geq |x| \geq x = \mathrm{Re}\, z, \qquad |z| \geq |y| \geq y = \mathrm{Im}\, z.$$

(c) $(z^*)^* = (x - iy)^* = x + iy = z$.
(d) If z is real, then $y = 0$, and hence $z^* = x - 0i = x = z$.

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1.2.1 Useful properties

1. $|z| = |z^*|$.
2. $z \cdot z^* = |z|^2$ (multiplication by the conjugate gives a real number).
3. $|zw| = |z|\,|w|$.
4. $(zw)^* = z^* w^*$.

EXERCISE 1.3

For $z \neq 0$, please prove that $(z^{-1})^* = (z^*)^{-1}$.

WORKED SOLUTION

Since $z \neq 0$, we also have $z^* \neq 0$ and $|z|^2 > 0$. The inverse formula gives

$$(z^{-1})^* = \left(\frac{z^*}{|z|^2}\right)^* = \frac{z}{|z|^2}, \qquad (z^*)^{-1} = \frac{(z^*)^*}{|z^*|^2} = \frac{z}{|z|^2}.$$

Both sides equal the same expression, so $(z^{-1})^* = (z^*)^{-1}$.

1.3 Polar form

1.3.1 Converting between Cartesian and polar forms

DEFINITION 1.4

For every $z = x + iy \neq 0$, there are $r = |z| > 0$ and an angle θ such that

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad x = r\cos\theta, \quad y = r\sin\theta.$$

Conversely, $r = \sqrt{x^2 + y^2}$. If $x \neq 0$, then $\tan\theta = y/x$, but this ratio alone does not determine the quadrant. **Remarks:**

- The angle is determined up to adding $2\pi k$, where $k \in \mathbb{Z}$. For $z = 0$, we have $r = 0$, and no argument is defined.
- All angles in exponents $e^{i\theta}$ are measured in radians. Degrees are used only for numerical illustration.

EXERCISE 1.4

Please convert $z = 5+5i$ and $w = -2+2i$ to polar form. Please write $a = 10e^{2i\pi/3}$ in Cartesian form. Please choose arguments in $[0, 2\pi)$.

WORKED SOLUTION

We have $|z| = \sqrt{25+25} = 5\sqrt{2}$, so $z = 5\sqrt{2}e^{i\pi/4}$. Similarly $|w| = \sqrt{4+4} = 2\sqrt{2}$. Its second-quadrant position gives $w = 2\sqrt{2}e^{3i\pi/4}$. Euler's identity gives

$$a = 10\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 10\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -5 + 5\sqrt{3}i.$$

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1.3.2 Arithmetic operations

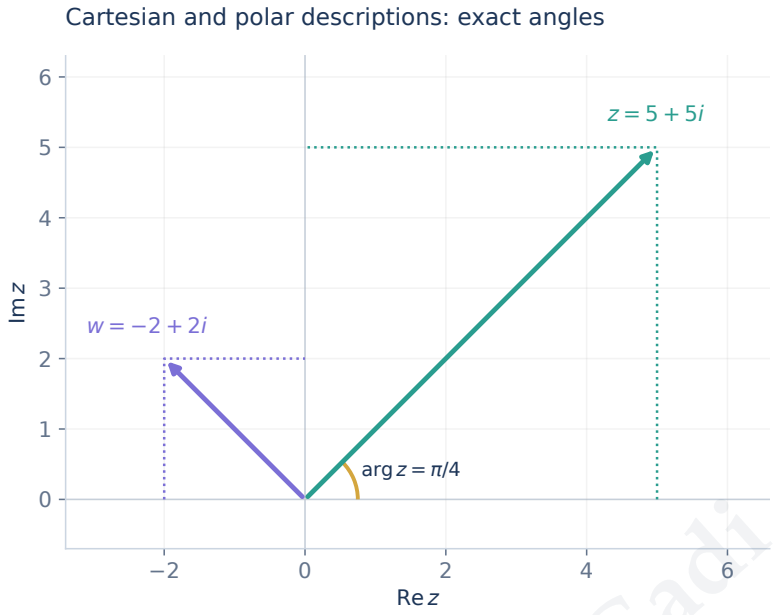


Figure 1.1. Cartesian and polar forms in the same plane: the modulus is the distance from the origin, and the argument is the direction angle.

DEFINITION 1.5

Multiplication: $(r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2\, e^{i(\theta_1+\theta_2)}$.

Division (when the denominator is nonzero): $\dfrac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \dfrac{r_1}{r_2}\, e^{i(\theta_1-\theta_2)}$.

Conjugation: $(re^{i\theta})^* = r\, e^{-i\theta}$.

Periodicity: $re^{i(\theta+2\pi k)} = re^{i\theta}$, $re^{i(\theta+\pi)} = -re^{i\theta}$.

EXERCISE 1.5

Let $z = 2+2i$ and $w = -3+3i$. Please calculate zw and w/z using polar and Cartesian forms. Please check their moduli as well.

WORKED SOLUTION

Write $z = 2\sqrt{2}e^{i\pi/4}$ and $w = 3\sqrt{2}e^{3i\pi/4}$. Then

$$zw = 12e^{i\pi} = -12, \qquad \frac{w}{z} = \frac{3}{2}e^{i\pi/2} = \frac{3}{2}i.$$

In Cartesian form,

$$(2+2i)(-3+3i) = -6+6i-6i+6i^2 = -12,$$

$$\frac{-3+3i}{2+2i} = \frac{(-3+3i)(2-2i)}{8} = \frac{-6+12i+6}{8} = \frac{3}{2}i.$$

The moduli are $|z||w| = 2\sqrt{2}\cdot 3\sqrt{2} = 12$ and $|w|/|z| = 3/2$, as expected.

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1.3.3 Powers

EXERCISE 1.6

Please calculate $(1-i)^{16}$ using polar form. Please verify the result using $(1-i)^4$.

WORKED SOLUTION

Since $1-i=\sqrt{2}e^{-i\pi/4}$,

$$(1-i)^{16}=(\sqrt{2})^{16}e^{-4\pi i}=2^8=256.$$

For a check, $(1-i)^2=-2i$ and $(1-i)^4=(-2i)^2=-4$, so $(1-i)^{16}=(-4)^4=256$.

1.3.4 Complex roots

DEFINITION 1.6

Let $0\neq z\in\mathbb{C}$, and let $n\geq 1$ be an integer. The equation $w^n=z=re^{i\theta}$ has n distinct roots:

$$w_j=\sqrt[n]{r}\,e^{i\frac{\theta+2\pi j}{n}},\quad j=0,1,\ldots,n-1.$$

Geometric meaning: the roots lie on the circle of radius $\sqrt[n]{r}$ at equal angular intervals of $2\pi/n$. For $n\geq 3$, they are the vertices of a regular polygon; for $n=2$, they are two opposite points, and for $n=1$, there is one point.

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EXERCISE 1.7

Please find all solutions of $z^3=8i$.

WORKED SOLUTION

Write $8i=8e^{i\pi/2}$. The three solutions are

$$z_j=2e^{i(\pi/6+2\pi j/3)},\qquad j=0,1,2.$$

Their angles are $\pi/6, 5\pi/6, 3\pi/2$, with radius 2. In Cartesian form they are $\sqrt{3}+i, -\sqrt{3}+i, -2i$. Cubing any of them gives $8e^{i\pi/2}=8i$.

EXERCISE 1.8

Let $n\geq 2$. Please find all nth roots of unity and please prove that their sum is zero. What happens when $n=1$?

WORKED SOLUTION

The roots are $w_j=e^{i\frac{2\pi j}{n}}$, $j=0,\ldots,n-1$. Put $w=e^{i\frac{2\pi}{n}}$. Then $w_j=w^j$, and therefore

$$\sum_{j=0}^{n-1}w^j=\frac{w^n-1}{w-1}=0,$$

because $w^n=1$ and $w\neq 1$ when $n\geq 2$. For $n=1$, the only root is 1, so the sum is 1, rather than zero.

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EXERCISE 1.9

(At home) Consider $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0$, with real coefficients. Please prove that if z is a solution, then z^* is also a solution.

WORKED SOLUTION

Use the identities $(z_1 + \cdots + z_m)^* = z_1^* + \cdots + z_m^*$ and $(z_1 \cdots z_m)^* = z_1^* \cdots z_m^*$, together with $a^* = a$ for real a. If $a_0 + a_1 z + \cdots + a_n z^n = 0$, conjugating gives $a_0 + a_1 z^* + \cdots + a_n (z^*)^n = 0$. Thus z^* is a solution.

1.4 Further properties of complex numbers

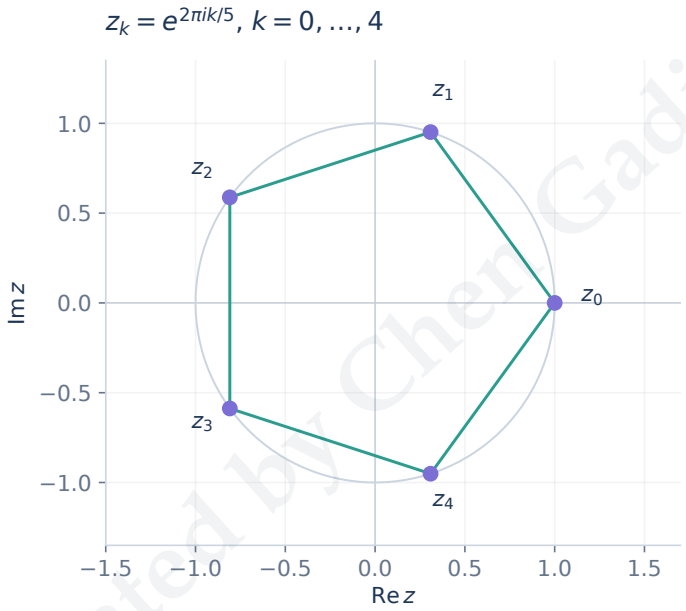


Figure 1.2. The roots of unity are equally spaced around the unit circle; their sum is zero whenever their order is greater than one.

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1.4.1 Multiplication by i

Multiplication by i rotates a complex number through $\frac{\pi}{2}$ counterclockwise.

EXERCISE 1.10

Please prove that multiplication by i gives a rotation through $\frac{\pi}{2}$.

WORKED SOLUTION

Cartesian representation:
Let $z = a + bi$. Multiplication by i gives

$$z \cdot i = (a + bi)i = -b + ai.$$

The vector (a, b) becomes $(-b, a)$. This is exactly a counterclockwise rotation through $\pi/2$ about the origin: the new x coordinate is $-b$, and the new y coordinate is a.

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EXERCISE 1.10 CONTINUED

Polar representation:
For $z \neq 0$, write $z = re^{i\theta}$, where $r = |z|$ and θ is an angle from the positive x axis to z. Then

$$z \cdot i = re^{i\theta} \cdot i = re^{i\theta} \cdot e^{i\frac{\pi}{2}} = re^{i(\theta+\frac{\pi}{2})}.$$

Multiplication by i adds $\frac{\pi}{2}$ to θ, so z rotates through $\frac{\pi}{2}$ about the origin, as required. The origin is fixed, since $0 \cdot i = 0$.

1.4.2 Division in polar form

DEFINITION 1.7

If $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, with $r_2 \neq 0$, then

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

EXERCISE 1.11

Please calculate z_1/z_2 for $z_1 = 9e^{5i\pi/6}$ and $z_2 = 3e^{i\pi/3}$.

WORKED SOLUTION

Divide the moduli and subtract the angles:

$$\frac{z_1}{z_2} = \frac{9}{3} e^{i(5\pi/6 - \pi/3)} = 3e^{i\pi/2} = 3i.$$

EXERCISE 1.12

Let $z_1, z_2 \neq 0$. Please prove that an argument of z_1/z_2 is the difference of arguments of z_1 and z_2, up to an integer multiple of 2π.

WORKED SOLUTION

Write $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$, where $r_1, r_2 > 0$. Division gives

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

Thus $\theta_1 - \theta_2$ is an argument of the quotient. If an argument in a prescribed interval is required, add a suitable integer multiple of 2π.

1.4.3 Choosing an argument in a fixed interval

In these notes, $\mathrm{Arg}(z)$ denotes the unique argument in $[0, 2\pi)$, for $z \neq 0$.

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RECALL AND TOOLS

The equalities $\text{Arg}(z_1 \cdot z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$ and $\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2)$ do not always hold, because the expression on the right may lie outside $[0, 2\pi)$.

EXERCISE 1.13

Please calculate $\text{Arg}(z_1 z_2)$ for $z_1 = e^{7i\pi/6}$ and $z_2 = e^{11i\pi/6}$.

WORKED SOLUTION

The product is

$$z_1 z_2 = e^{i(7\pi/6+11\pi/6)} = e^{3\pi i} = -1.$$

Since 3π lies outside $[0, 2\pi)$,

$$\text{Arg}(z_1 z_2) = \pi, \qquad \text{Arg}(z_1) + \text{Arg}(z_2) = 3\pi.$$

The two values differ by 2π.

1.4.4 Calculating arguments and identifying quadrants

DEFINITION 1.8

For a complex number $z = a + bi \neq 0$ with $a \neq 0$, every argument θ satisfies

$$\tan(\theta) = \frac{b}{a}.$$

To determine θ precisely, take the quadrant into account:

- If $a > 0, b > 0$, then $\theta = \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b > 0$, add π to obtain an angle in the second quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a < 0, b < 0$, add π for the third quadrant: $\theta = \pi + \arctan\left(\frac{b}{a}\right)$.
- If $a > 0, b < 0$, add 2π to obtain a positive angle in the fourth quadrant: $\theta = 2\pi + \arctan\left(\frac{b}{a}\right)$.

If $b = 0, a > 0$, then $\text{Arg}(z) = 0$; if $b = 0, a < 0$, then $\text{Arg}(z) = \pi$. If $a = 0$, the angle depends only on b:

- If $b > 0$, then $\theta = \frac{\pi}{2}$.
- If $b < 0$, then $\theta = \frac{3\pi}{2}$.

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EXERCISE 1.14

Please find the argument in $[0, 2\pi)$ of $z = -\sqrt{3} + i$.

WORKED SOLUTION

The number lies in the second quadrant, and

$$\tan\theta = \frac{1}{-\sqrt{3}} = -\frac{1}{\sqrt{3}}.$$

Its reference angle is $\pi/6$, so $\theta = \pi - \pi/6 = 5\pi/6$.

EXERCISE 1.15

Please find the argument in $[0, 2\pi)$ of $z = -6 - 2\sqrt{3}i$.

WORKED SOLUTION

The number lies in the third quadrant, and

$$\tan\theta = \frac{-2\sqrt{3}}{-6} = \frac{1}{\sqrt{3}}.$$

Its reference angle is $\pi/6$, so $\theta = \pi + \pi/6 = 7\pi/6$.

1.5 The induction principle

DEFINITION 1.9

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 1.16

Please prove by induction that $1 + 3 + \cdots + (2n-1) = n^2$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, the identity is $1 = 1^2$. Suppose that for $k \geq 1$,

$$\sum_{j=1}^{k}(2j-1) = k^2.$$

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EXERCISE 1.16 CONTINUED

Adding the next term gives

$$\sum_{j=1}^{k+1}(2j-1) = k^2 + (2k+1) = (k+1)^2.$$

Induction proves the claim for every $n \geq 1$.

EXERCISE 1.17

Please prove by induction that $11 \mid (12^n - 1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $12 - 1 = 11$. Suppose $12^k - 1 = 11m$ for some $m \in \mathbb{Z}$. Then

$$12^{k+1} - 1 = 12(12^k - 1) + 11 = 12 \cdot 11m + 11 = 11(12m + 1).$$

This completes the induction step. As a check, $12 \equiv 1 \pmod{11}$.

EXERCISE 1.18

Please prove by induction that $3^n \geq n^2$ for every $n \geq 2$.

WORKED SOLUTION

For $n = 2$, $3^2 = 9 \geq 4 = 2^2$. Suppose $3^k \geq k^2$ for $k \geq 2$. Then

$$3^{k+1} \geq 3k^2.$$

Moreover,

$$3k^2 - (k+1)^2 = 2k(k-1) - 1 \geq 3 > 0,$$

so $3^{k+1} \geq (k+1)^2$. Induction proves the claim for all $n \geq 2$.

1.6 Supplementary exercises: geometry and signal calibration

EXERCISE 1.19

Please find and please describe geometrically all $z \in \mathbb{C}$ satisfying $|z-2| = |z+2|$. Please find which of them also satisfy $|z| = 5$.

WORKED SOLUTION

Writing $z = x + iy$ and squaring gives

$$(x-2)^2 + y^2 = (x+2)^2 + y^2 \iff x^2 - 4x + 4 = x^2 + 4x + 4 \iff x = 0.$$

This is the imaginary axis, the perpendicular bisector of the segment joining -2 and 2. The extra condition is $y^2 = 25$, giving $z = 5i$ or $z = -5i$.

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EXERCISE 1.20

Let $T(z) = (2 - 2i)z$. Please determine its effect on modulus and angle. Please calculate $T(1 + i)$ and please find an input whose output is $8i$.

WORKED SOLUTION

Since $2 - 2i = 2\sqrt{2}e^{-i\pi/4}$, the map multiplies moduli by $2\sqrt{2}$ and rotates clockwise through $\pi/4$. The origin stays fixed. Directly,

$$T(1 + i) = (2 - 2i)(1 + i) = 2 + 2i - 2i - 2i^2 = 4.$$

To obtain $8i$, divide by the nonzero multiplier:

$$z = \frac{8i}{2 - 2i} = \frac{8i(2 + 2i)}{8} = -2 + 2i.$$

Check: $(2 - 2i)(-2 + 2i) = -4 + 8i + 4 = 8i$.

Additional practice

EXERCISE 1.21 FOUNDATION

Let $z = 5 - 12i$. Please calculate $|z|$, z^* and z^{-1}. Please solve $zw = 22 - 19i$ and please verify by direct multiplication.

WORKED SOLUTION

We have $|z| = \sqrt{25 + 144} = 13$, so $z^* = 5 + 12i$ and $z^{-1} = (5 + 12i)/169$. Hence

$$w = \frac{22 - 19i}{5 - 12i} = \frac{(22 - 19i)(5 + 12i)}{169} = \frac{338 + 169i}{169} = 2 + i.$$

Check: $(5 - 12i)(2 + i) = 10 + 5i - 24i - 12i^2 = 22 - 19i$. The solution is unique because $z \neq 0$.

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EXERCISE 1.22 CHALLENGE

Let $z \in \mathbb{C} \setminus \{-1\}$. Please prove that $(z-1)/(z+1)$ is purely imaginary if and only if $|z| = 1$. Please solve $(z-1)/(z+1) = 3i$ and please check the geometric condition.

WORKED SOLUTION

Multiply by the conjugate of the denominator. Since $z \neq -1$, $|z+1|^2 > 0$ and

$$\frac{z-1}{z+1} = \frac{|z|^2 - 1 + z - z^*}{|z+1|^2}.$$

The term $z - z^*$ is purely imaginary, so

$$\operatorname{Re}\left(\frac{z-1}{z+1}\right) = \frac{|z|^2 - 1}{|z+1|^2}.$$

This vanishes if and only if $|z| = 1$, including $z = 1$ where the quotient is zero. The given equation yields $z - 1 = 3iz + 3i$, hence

$$z = \frac{1+3i}{1-3i} = \frac{(1+3i)^2}{10} = \frac{-4+3i}{5}.$$

This differs from -1 and has modulus $\sqrt{16+9}/5 = 1$. Finally,

$$\frac{z-1}{z+1} = \frac{-9+3i}{1+3i} = 3i.$$

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EXERCISE 1.23 CHALLENGE

Let $a, b, c \in \mathbb{C}$ be pairwise distinct. Please prove that their points form an equilateral triangle if and only if

$$a^2 + b^2 + c^2 = ab + bc + ca.$$

WORKED SOLUTION

Since $b \neq a$, put $w = (c-a)/(b-a)$. The map $z \mapsto (z-a)/(b-a)$ sends the points to $0, 1, w$ and multiplies every distance by $1/|b-a|$, so it preserves equilateral triangles. Substituting $c = a + (b-a)w$ gives

$$a^2 + b^2 + c^2 - ab - bc - ca = (b-a)^2(w^2 - w + 1).$$

Thus the identity is equivalent to $w^2 - w + 1 = 0$, whose roots are $w = 1/2 \pm i\sqrt{3}/2$. Both satisfy $|w| = |w-1| = 1$. Conversely, if $0, 1, w$ form an equilateral triangle and $w = u + iv$, then

$$u^2 + v^2 = 1, \qquad (u-1)^2 + v^2 = 1.$$

Subtracting yields $u = 1/2$ and then $v^2 = 3/4$. These are exactly the two roots above, so the identity follows.

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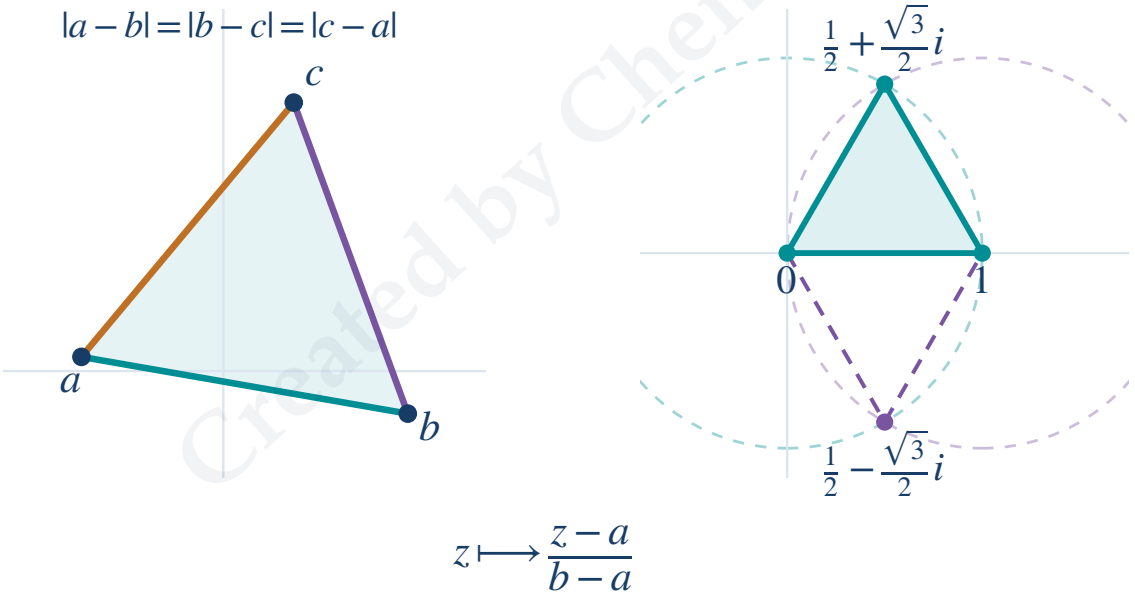


Figure 1.3. The map $z \mapsto (z-a)/(b-a)$ sends the vertices to $0, 1, w$. The two possible third vertices are $w = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$.

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EXERCISE 1.24 CHALLENGE

Let $a, b, c \in \mathbb{C}$ satisfy $|a| = |b| = |c| = 1$. Please prove that $|a + b + c| = 1$ if and only if at least one of $a + b, a + c, b + c$ is zero.

WORKED SOLUTION

Put $s = a + b + c$. Since each of a, b, c has modulus 1, its conjugate equals its reciprocal. Thus

$$ab + ac + bc = abc\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = abc\,\overline{s}.$$

Expanding gives

$$\begin{aligned}(a + b)(a + c)(b + c) &= (a + b + c)(ab + ac + bc) - abc \\ &= abc(s\overline{s} - 1) = abc(|s|^2 - 1).\end{aligned}$$

If $|s| = 1$, the product on the left is zero, so at least one factor is zero. Conversely, if for example $a + b = 0$, then $s = c$ and $|s| = 1$. The other two pairs give the same conclusion.

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